Rodrigo G. Freundt

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Summary

- Research-oriented engineer and PhD candidate in Astronomy and Space Sciences with +7 years of experience building and integrating astronomical instrumentation. Excellent technical writing, project management and mentoring skills. Experience working in large collaborations with diverse teams of scientists and engineers.

Education

Cornelll University Ithaca, NY
Ph.D. in Astronomy and Space Sciences. 2021–Present

- Thesis: "The Epoch of Reionization Spectrometer (EoR-Spec): Development and Early Science." (in preparation). Expected completion: July 2027.

Pontifical Catholic University of Peru (PUCP) Lima, Peru
B.Sc. in Electronics Engineering. 2011–2016

- Thesis: "Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU".

Experience

CCAT collaboration Ithaca, NY.
Graduate Research Assistant. Advisors: Prof. Gordon Stacey and Prof. Michael Niemack 2021 –Present

- Leading the design, fabrication, testing and deployment (early 2027) of EoR-Spec, a line intensity mapping experiment for the Fred Young Submillimeter Telescope (FYST).
- Designed and integrated a cryogenic readout harness (capable of reading out >25,000 kinetic inductance detectors) for Prime-Cam, a modular receiver for FYST.

Atacama Cosmology Telescope San Pedro de Atacama, Chile.
Site Engineer. Advisors: Prof. Suzanne Staggs and Prof. Mark Devlin. Oct 2019 –Jul 2021

- Ensured the continuous operation of complex astronomical instrumentation at extreme altitude (17,000 ft). In charge of regular maintenance, troubleshooting and overhaul of electrical, cryogenic and mechanical systems.
- Performed calibration procedures, such as the alignment of the receiver with the primary and secondary reflectors through high-precision photogrammetry ($\sim 10 \mu$m). Carried out seasonal passband characterization via Fourier Transform Spectroscopy.
- Helped in the deployment of the last detector array upgrade (27/39 GHz) for the ACTPol receiver.

Management Solutions Lima, Peru / Madrid, Spain.
Assistant Business Consultant Sep 2018 –Jul 2019

- Developed bank stress test software for a major local bank in Lima, Peru. Developed interactive data visualization dashboards for a leading financial institution in Madrid, Spain.

Institute for Radio Astronomy - PUCP Lima, Peru.
Research Engineer (Intern until Apr 2017) Advisor: Prof. Jorge Heraud Mar 2015 –Aug 2018

- Worked in the construction of an 8-meter radio telescope (RT-8) intended for teaching and outreach. Designed custom hardware and software for the telescope's pointing system.
- Developed custom software and firmware for an heterogenous digital spectrometer.
- Assisted in the characterization of a 1.42 GHz feed antenna.

Skills

Optical, vacuum and cryogenic system design. Measurement and control automation. Collaborative mindset.

- **Programming:** Python, C/C++, CUDA, LabVIEW.
- **CAD/Simulation** Solidworks, KiCAD, Ansys Zemax Optics Studio.
- **Tools:** LaTeX, Git, Docker.
- **Languages:** English (fluent), Spanish (native).

Awards

- CASPER Workshop, Harvard & Smithsonian Center for Astrophysics (CfA) — Aug 2019
  Merit-based fee waiver and travel grant (invited, could not attend). Cambridge, MA.
- Summer School of Astronomical Instrumentation, Dunlap Institute, UofT — Jul 2019
  Merit-based tuition waiver and travel grant. Toronto, ON.
- URSI Young Scientist Award — Aug 2017
  Merit-based registration fee waiver and travel grant to attend URSI 2017, Montreal, QC.
- Beca de Estímulo Académico Solidario (BEAS) — 2013–2016
  Fully-funded undergraduate scholarship for outstanding academic performance and financial need.

Teaching

- **Teaching Assistant** at Cornell University. — Fall 2022, 2024, 2025, 2026
  Multiwavelenght astronomical techniques (ASTRO 4410)
  Advisors: Prof. Jim Cordes and Prof. Gordon Stacey
- **Teaching Assistant** at Cornell University — Spring 2023
  Our Solar System (ASTRO 1102)
- **Teaching Assistant** at PUCP — Aug 2017 - Dec 2017
  Space Science and Engineering (ING 306)

Academic Presentations

Talk: SPIE Astronomical Telescopes + Instrumentation 2026 — Copenhagen, Denmark
CCAT: The Epoch of Reionization Spectrometer (EoR-Spec) instrument status — Jul 2026

Talk: SPIE Astronomical Telescopes + Instrumentation 2024 — Yokohama, Japan
CCAT: A status update on the EoR-Spec instrument module for Prime-Cam — Jun 2024

Group talk: LIM 2024 — Urbana, IL.
CCAT: The Deep Spectroscopic Survey with The Epoch of Reionization Spectrometer — Jun 2024

Talk 243rd meeting of the American Astronomical Society — New Orleans, LA.
CCAT: The Epoch of Reionization Spectrometer. — Jan 2024

Publications

[1] B. Keller, **R. Freundt**, J. R. Burgoyne, S. Chapman, S. Choi, C. J. Duell, C. Groppi, C. Humphreys, L. T. Lin, A. Middleton, M. D. Niemack, D. Patel, E. Vavagiakis, S. Walker, Y. Wang, and R. Xie, "Ccat: Flexible stripline circuits for large-format kinetic inductance detector (kid) array readout", *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 1–6, 2026.

[2] **R. Freundt**, Y. Li, D. Henke, J. Austermann, J. R. Burgoyne, S. Chapman, S. K. Choi, C. J. Duell, Z. Huber, M. Niemack, T. Nikola, L. Lin, D. A. Riechers, G. Stacey, A. K. Vaskuri, E. M. Vavagiakis, J. Wheeler, and B. Zou, "CCAT: A status update on the EoR-Spec instrument module for Prime-Cam", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 343–352.

[3] **R. Freundt** and J. A. Heraud, "Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU", in *2017 XXXIInd General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS)*, Aug. 2017, pp. 1–4.

Collaboration Publications

[4] Lin, L. T. et al. (including **R. G. Freundt**), "Ccat: Mod-cam cryogenic performance and its impact on 280-ghz kid array noise", *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 1–7, 2026, ISSN: 2378-7074.

[5] Patel, D. A. et al. (including **R. G. Freundt**), "Ccat: Design, implementation, and testing of a system to read out over 10000 280 ghz kids using rfsoc electronics", *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 2 503 607–2 503 607, 2026, ISSN: 2378-7074.

[6] Calabrese, E. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 constraints on extended cosmological models", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 063, Nov. 2025, ISSN: 1475-7516.

[7] Louis, T. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 power spectra, likelihoods and lcdm parameters", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 062, Nov. 2025, ISSN: 1475-7516.

[8] Middleton, A. et al. (including **R. G. Freundt**), "Ccat: Led mapping and characterization of the 280 ghz tin kid array", *IEEE Transactions on Applied Superconductivity*, vol. 35, no. 5, pp. 1–4, 2025.

[9] Naess, S. et al. (including **R. G. Freundt**), "The atacama cosmology telescope: Dr6 maps", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 11, p. 061, Nov. 2025, ISSN: 1475-7516.

[10] Vaskuri, A. K. et al. (including **R. G. Freundt**), "280-GHz aluminum MKID arrays for the Fred Young Submillimeter Telescope", *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 11, 026005, p. 026 005, Apr. 2025.

[11] Coulton, W. et al. (including **R. G. Freundt**), "Atacama cosmology telescope: High-resolution component-separated maps across one third of the sky", *Physical Review D*, vol. 109, no. 6, Mar. 2024, ISSN: 2470-0029.

[12] B. Zou, J. R. Bond, V. Butler, **R. Freundt**, Z. B. Huber, M. D. Niemack, T. Nikola, G. J. Stacey, and E. M. Vavagiakis, "CCAT: Characterizing the Fabry-Pérot interferometer mirrors and mount for the epoch of reionization spectrometer", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 909–916.

[13] Choi, S. K. et al. (including **R. G. Freundt**), "Ccat-prime: Characterization of the first 280 ghz mkid array for prime-cam", *Journal of Low Temperature Physics*, vol. 209, no. 5-6, pp. 849–856, Aug. 2022, ISSN: 1573-7357.

[14] Collaboration, C.-P. et al. (including **R. G. Freundt**), "CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope", *The Astrophysical Journal Supplement Series*, vol. 264, no. 1, p. 7, Dec. 2022, et al. including R. G. Freundt, ISSN: 0067-0049.

[15] T. Nikola, S. K. Choi, C. J. Duell, R. G. Freundt, Z. B. Huber, Y. Li, K. Malavalli, M. Niemack, K. M. Rossi, G. J. Stacey, E. M. Vavagiakis, B. Zou, N. F. Cothard, J. Austermann, J. D. Wheeler, J. Gao, M. R. Vissers, J. Hubmayr, J. Beall, and J. Ullom, "CCAT-prime: The epoch reionization spectrometer for primce-cam on FYST", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI*, vol. 12190, SPIE, Aug. 31, 2022, pp. 247–262.

[16] B. Zou, S. K. Choi, N. F. Cothard, **R. Freundt**, Z. B. Huber, Y. Li, M. D. Niemack, T. Nikola, D. A. Riechers, K. M. Rossi, G. J. Stacey, and E. M. Vavagiakis, "CCAT-prime: The design and characterization of the silicon mirrors for the Fabry-Perot interferometer in the Epoch of reionization spectrometer", in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI*, vol. 12190, SPIE, Aug. 31, 2022, pp. 982–994.